

Chapter 1

TT

1.1 SYS-0

The alphabet is listed as follows.

- ◊ $V = \{v_i\}_{i \geq 0} \cup \{A_i\}_{i \geq 0}$;
- ◊ C is the set of constant symbols;
- ◊ $B = \{\Pi, \lambda, \Sigma, \dots\}$ is the set of binders;
- ◊ $U = \{\text{Type}_i\}_{i \geq 0}$ is the set of universes;
- ◊ $P = \{(\cdot), \cdot, \equiv\}$ is the set of punctuation.

All of these sets are at most countable.

The whole formulas we use is contained by set

$$\mathbf{F} = V|C|U|B(V : \mathbf{F})\mathbf{F}|(\mathbf{F}\mathbf{F})|(\mathbf{C}\mathbf{F}^*).$$

Notation 1.1.1 Here we announce some commonly used notations for avoiding ambiguity.

- ◊ $\mathbf{a} = a_0 a_1 \dots a_m := (a_0, a_1, \dots, a_m)$;
- ◊ $f(\mathbf{x}) := f$ where $f \in \mathbf{F}$ and $\text{Fr } f \subset \{x_0, \dots, x_m\}$;
- ◊ $(f\mathbf{x})$ refers $(\dots((fx_0)x_1)\dots)$, note that this may differ the notation $(fx_0x_1\dots x_m)$, which is equal to our first definition;
- ◊ for any $b \in B$, $b(\mathbf{v} : \mathbf{A})N := b(v_0 : A_0)b(v_1 : A_1)\dots b(v_s : A_s)N$;
- ◊ $\mathbf{v} : \mathbf{A}$ refers

$$v_0 : A_0 \quad v_1 : A_1 \quad \dots \quad v_s : A_s;$$
- ◊ $M \rightarrow N := \Pi(v : M)N$, in which $x \notin \text{Fr } N$;
- ◊ to distinguish between terms in $\mathbf{F}$ and constants in C , we use a mono font for constants.

Definition 1.1.2 A definition D has the form

$$\mathbf{c}(\mathbf{v} : \mathbf{A}) \equiv d(\mathbf{v}) : B,$$

where $v \in V$, $c \in C$ and all of $A, B, d(\mathbf{v})$ are in $\mathbf{F}$.

Definition 1.1.3 A sequent has the form

$$D \quad \mathbf{v} : \mathbf{A} \quad M : A$$

where $M : A$ is called a judgement, and both M and A are in $\mathbf{F}$. The set of all sequents is denoted by Sqt .

Legal judgements consist a subset of $\mathbf{F} : \mathbf{F}$, which is derived from a series of **rules**. In our SYS-0, we first introduce the following rules. Each of these rules can be viewed as a relation on $\text{Sqt}^* \times \text{Sqt}$.

Sort

If $m < n$, then

$$_ _ \text{Type}_m : \text{Type}_n$$

Var

If $v \in V \setminus (\Gamma \cap V)$, $s \in U$, then

$$\begin{array}{l} \mathbf{D} \quad \Gamma \quad A : s \\ \mathbf{D} \quad \Gamma \quad v : A \quad v : A \end{array}$$

Weak

If $v \in V \setminus (\Gamma \cap V)$, then

$$\begin{array}{l} \mathbf{D} \quad \Gamma \quad M : A \\ \mathbf{D} \quad \Gamma \quad B : s \\ \mathbf{D} \quad \Gamma \quad v : B \quad M : A \end{array}$$

1.2 SYS-1

我们通过引入新符号, 增加新规则和关系来增加新的项和类型, 这个过程大致遵循一个固定的 workflow: 1. Formation: 引入新规则, 让系统在指定条件下可以导出新类型; 2. Constuction: 引入新规则, 让系统在指定条件下可以导出新项, 并且这些项都具有新类型; 3. Elimination: 引入新规则, 告知系统新项应该如何使用; 4. Computaiton: 引入 F 上的新关系, 告知系统新系统中某些表达和旧系统中的某些表达具有同样的语义.

现在我们将拓展 SYS-0, 增加 λ 项和 Π 类型.

SYS-1: Form

If $s_1 \leq s_2$, then

$$\begin{array}{l} \mathbf{D} \quad \Gamma \quad A : s_1 \\ \mathbf{D} \quad \Gamma \quad v : A \quad B : s_2 \\ \mathbf{D} \quad \Gamma \quad \Pi(v : A)B : s_2 \end{array}$$